When Hiring Becomes Agent-Mediated:

Evaluating Access and Recurrence in Two-Agent Résumé Screening

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Key takeaway

Two procedures can advance similar numbers of applications, but not the same applications. Some procedure-specific selections do not recur when the pipeline is re-run.

Why this matters

Résumé screening is the first hiring gate, and LLM-based screening is commonly **one model call**. As employers and candidates adopt agents, it can become an **exchange** between employer-side and candidate-side agents. A PASS is not a hire, but it **controls access to human review**.

Question: does two-agent screening change *which* applications pass, and if so, how?

Controlled testbed

200	200	200
Strong	Borderline	Weak

200 LinkedIn postings × 3 constructed résumés = **600 pairs** (LLM-free scorer: 600/600 agree). Models: **GPT-5.5**, **Claude Opus 4.7**. Runs A–C: both procedures 3× on a **191-pair borderline pool** (9 gated pairs excluded); D–I: fresh re-runs of selected cases.

Two screening procedures

One-call *compensatory*

Joint access — one model sees posting **and** résumé → **1 model call** — emits employer- and candidate-perspective scores → **Mean ≥ .75 → PASS** — a high score on one side can offset the other

Exactly **1 call** per decision; no pre-screen gate.

Two-agent *non-compensatory*

Skill-overlap gate — <10% of normalized required-skill names in résumé skill list → FAIL, no call (203/600) → **Employer ↔ Candidate** — each sees only its own document; evidence crosses via dialogue → **Rule layer** — .75 commit, .65 floor; PASS from round 3; soft cap 8, max 12 rounds

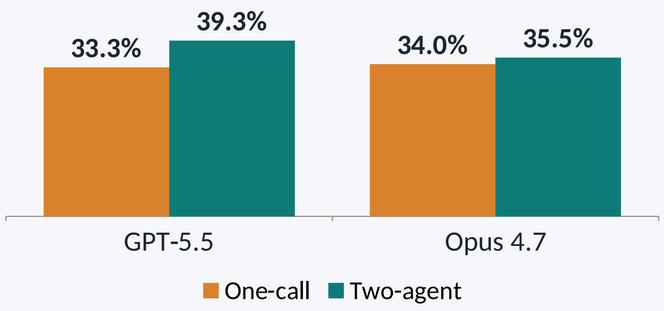
Median **8**, mean **9.31** calls per decision on the borderline pool.

No new applicant information enters the exchange; we compare **complete procedures**.

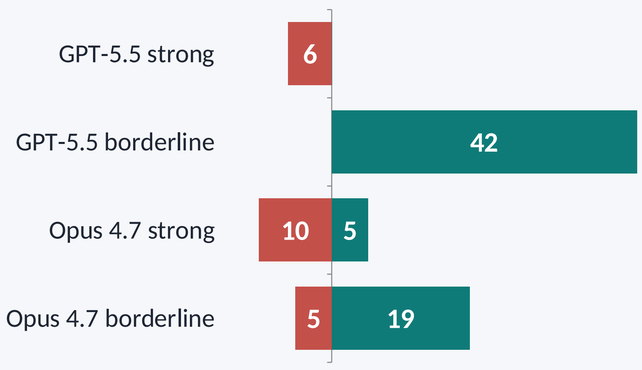
VOLUME & DIRECTION

1 Access changes in both directions

Run-A PASS rate, all 600 pairs



The small net change hides decisions moving both ways: **removed** (one-call PASS → two-agent FAIL) vs. **added**.

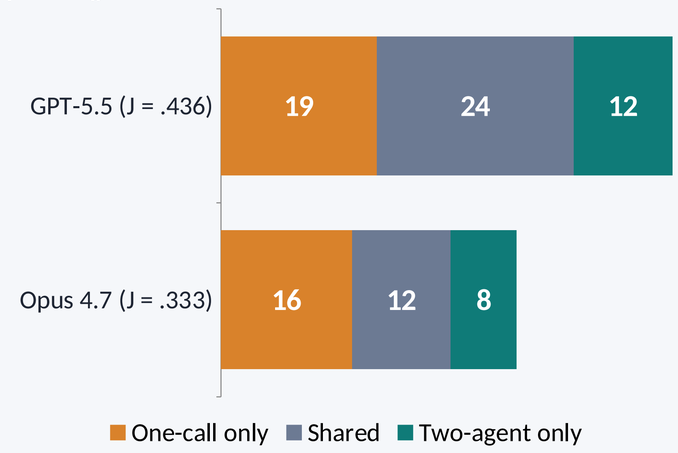


A uniformly lower one-call cutoff can only *add* passes, so removals rule out **simple relaxation**.

MEMBERSHIP

2 Similar volume, different applications

One-call scores re-read at the cutoff closest to two-agent pass volume (.590 / .575). Applications passing **all 3 runs**:



"Only" = 3/3 PASS under one procedure, <3/3 under the other.

No threshold on the stored one-call mean reconstructs the two-agent persistent set: best Jaccard across **all cutoffs** is .436 (GPT-5.5) and .361 (Opus 4.7).

Two-agent screening is **not another operating point** on the stored one-call mean-score ranking.

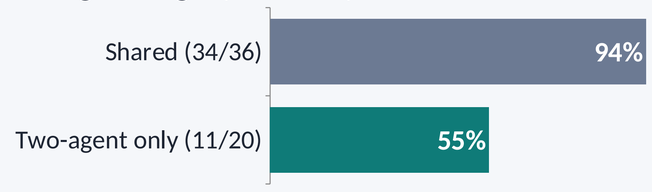
RECURRENCE

3 Lower recurrence among two-agent-only selections

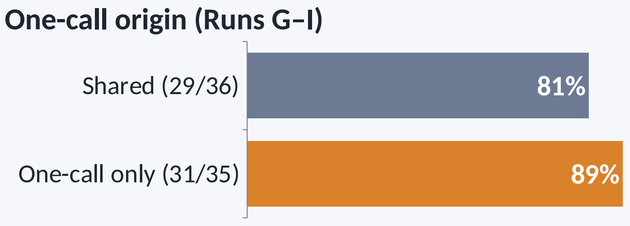
Fresh re-runs of discovery-selected cases: share again passing **3/3** under the originating procedure.

Pooled application–model cases; one-call at aligned cutoffs.

Two-agent origin (Runs D–F)



One-call origin (Runs G–I)



Per model: GPT-5.5 **24/24 vs 7/12** (Fisher p = .0021); Opus 4.7 **10/12 vs 4/8** (p = .161, uncertain).

Caveat. Overall recurrence is similar: 45/56 (80.4%) two-agent vs. 60/71 (84.5%) one-call. Selected cohorts may differ in latent difficulty; this is **not** evidence that two-agent screening is generally less reliable.

Evaluate the whole procedure: report all three levels

Volume — How many applications advance?

Membership — Which ones advance at aligned volume?

Recurrence — Do distinguishing selections survive re-execution?

IMPLICATION

The model matters. The procedure also determines who reaches human review and how reliably that access recurs.

Scope. Constructed résumés and public postings; no real applicants or hiring outcomes. We compare complete procedures and do not attribute differences to dialogue, role separation, compute, gating, or rules, nor claim accuracy or fairness. GPT-5.5 also generated the résumés.